

AMEYA SWAPNEEL KORE

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EDUCATION

Master of Science in Computer Science

Graduating May 2028

University of Southern California, Los Angeles, CA

Relevant coursework: Autonomous Cyber-Physical Systems, Analysis of Algorithms

Bachelor of Engineering in Computer Science and Engineering

Graduated April 2025

Vellore Institute of Technology, Vellore, India

CGPA: 9.63 (Top 0.5% in Department)

Relevant coursework: Data Structures and Algorithms, Operating Systems, Compiler Design, Cloud Computing, Control Systems

TECHNICAL SKILLS

Languages: Rust, Python, Java, C/C++, JavaScript, TypeScript, Kotlin, Dart, SQL

Systems & Tools: Docker, Kubernetes, AWS EC2, Microsoft Azure, Linux, RTOS, DevOps

EXPERIENCE

Motorq, Bengaluru, India: Software Development Engineer

Jun 2025 – Jul 2026

- Engineered internal AI remediation pipeline translating SonarQube static reports into remediated GitHub PRs.
- Built conditional workflow retry feature recovering 5% of onboarding operations, preventing revenue loss.
- Led OEM vehicle telemetry ingestion pipelines and executed zero-downtime database migrations for P0 services.

Daten & Wissen, Mumbai, India (Remote): Software Development Intern

Oct 2023 – Apr 2024

- Integrated ROI-based image filters into edge AI vision pipelines to optimize inference precision and throughput.

Florican Infosoft, Mumbai, India (Remote): Freelance Developer

Jan 2023 – Jun 2023

- Engineered multi-station chemical formulation software in Kotlin and Java SWT with real-time weighing scale sync.
- Built high-throughput AES encryption/decryption modules in Rust to secure proprietary recipe transmissions.

RESEARCH AND PROJECTS

Improving Model Accuracy in DDoS Prevention by Leveraging GANs

Published in IEEE

Lightweight GAN-based traffic synthesis addressing classifier bias in imbalanced IoT datasets

- Published in the IEEE International Conference on Intelligent Computing and Knowledge Extraction (ICICKE).
- Designed lightweight GAN generating synthetic benign traffic to balance skewed healthcare IoT datasets.
- Elevated DDoS classification accuracy from 97.13% to 99.61%, facilitating real-time edge deployment.

Autonomous RC Navigation System

Self-directed navigation stack on modified RC platform with custom embedded ESP32 RF bridge

- Reverse-engineered proprietary RF protocols and soldered an ESP32 bridge to the PCB for low-latency actuation.
- Built closed-loop orchestrator linking on-device VLM and YOLOE-26 to calibrate headings via micro-bursts.

MangaVerse – Distributed Comic Aggregation System

Unified high-throughput aggregation backend resolving multi-source data retrieval fragmentation

- Engineered multi-source scrapers and unified abstraction caching in Spring Boot, later re-architected in Rust.
- Reduced query latency by 50% on a 13GB dataset while substantially boosting high-load concurrency and throughput.

OPEN SOURCE & LEADERSHIP

- Open Source Contributions:** Implemented idiomatic Rust safe abstractions for light and deep sleep power management in esp-idf-hal, integrating low-power wake-up source bindings and hardware timer configurations.
- Google Developer Student Club (VIT):** Senior Core Member; organized hackathons; instructed workshop on JS runtime.
- Interests & Arts:** Trinity Level 4 (Keyboard), Indian Classical Vocals/Harmonium, Soccer, Table Tennis, Badminton.